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Enhancing engagement and collaborative creativity in health promotion education through co-creative game-based learning

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ABSTRACT

This study examines the impact of a co-creative game-based learning (GBL) intervention on student engagement and collaborative creativity in a Taiwanese undergraduate health promotion course. Responding to pedagogical reforms emphasising practice-oriented competencies, the course embedded a participatory task in which students collaboratively designed educational board games targeting elementary school audiences. Quantitative outcomes included learning engagement and collaborative creativity, measured through validated scales, alongside qualitative student reflections. The results exhibited significant improvements in both engagement and collaborative creativity, with the greatest gains observed among students with lower academic standing and motivation. Qualitative data highlighted students' enhanced capacity to translate abstract theory into accessible content, increased teamwork skills, and heightened professional identity awareness. The study contributes empirical evidence to the emerging field of participatory GBL, highlighting its effectiveness in promoting engagement, creativity, and equitable learning opportunities within competency-based health promotion education.

KEYWORDS

Co-creative game-based learning; health promotion education; student engagement; collaborative creativity; participatory pedagogy; public health

Introduction

Health promotion education bridges academic theory and community practice, requiring learners to develop both conceptual understanding and practical competencies such as teamwork, communication, and project implementation (MacKay et al., 2024). However,

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traditional lecture-based instruction often falls short in fostering critical thinking and transferable skills, limiting students' ability to apply knowledge in authentic contexts. Experiential and reflective learning, including game-based learning (GBL), offers a promising way to connect theory with practice and strengthen identity development (Pham et al., 2023).

GBL involves the use of games or game design principles to create instructional environments that integrate educational objectives, rules, and feedback to promote engagement and learning. It fosters active participation, reflection, and knowledge construction through play or design (Pan et al., 2021; Sailer & Homner, 2020). Rather than using games solely for entertainment, GBL aligns gameplay or game creation with pedagogical goals to enhance motivation and creativity (Sailer & Homner, 2020).

In Taiwan, pedagogical reform in health education has gained urgency since the enactment of the Public Health Specialists Act in 2020 (Yeh, 2024). Universities are now expected to cultivate practice-oriented competencies, including the ability to design and evaluate community-based initiatives. However, instruction remains lecture-driven, limiting opportunities for experiential and collaborative learning that mirror real-world practice.

Beyond national reforms, global frameworks highlight the importance of cultivating graduates with creativity, collaboration, communication, and critical thinking skills, aligning with the goals of health promotion education (Ho & Le, 2026; Thornhill-Miller et al., 2023).

In response to these needs, educators have sought innovative pedagogical models that integrate creativity and collaboration. Design-based learning (DBL) and co-creative GBL are promising approaches that position learners as active creators rather than passive players (Romero et al., 2019; Zaw Oo et al., 2024). Engaging students as game designers encourages them to analyse, reorganise, and apply theoretical concepts in meaningful contexts, thereby strengthening reflective thinking, facilitating the transfer of theory-to-practice, and fostering collaborative creativity (Casey & Hastie, 2011; Palmquist et al., 2024). This participatory approach aligns with health promotion principles, which emphasise empowerment, collaboration, and active engagement (Glăveanu & Beghetto, 2021; Thornhill-Miller et al., 2023), providing a relevant framework for preparing future public health professionals.

Studies illustrate that GBL enhances learners' motivation, engagement, and performance across disciplines (Jones et al., 2022; Zhang & Yu, 2022). These effects stem from its clear goals, feedback loops, and meaningful challenges that promote self-regulation. Within this framework, DBL represents an active, student-centred extension of GBL, in which learners design educational materials, apply theoretical concepts, and iteratively refine their work (Pan et al., 2021). This process fosters ownership and agency. Additionally, it supports students with lower achievement or confidence, offering them opportunities to demonstrate understanding through meaningful creation.

Although co-creative GBL has been explored in fields such as science and sustainability education (Heim & Holt, 2021; Mercer et al., 2017), it remains underrepresented in health education. Evidence from other disciplines suggests that engaging students as co-designers not only deepens motivation and self-efficacy but also enhances the perceived relevance of learning, linking abstract theory with authentic, practice-oriented application

(Romero et al., 2019). This gap highlights how co-creative, design-based GBL can foster engagement and collaborative creativity in health promotion education.

To address this, the present study implements a co-creative GBL approach in a university-level health promotion course, examining its effects on student engagement and collaborative creativity and exploring whether these outcomes vary by students' academic background or motivation. The study offers empirical insights to inform innovative, student-centred strategies in health promotion education.

Game-based learning and game design

GBL has been increasingly recognised as an effective pedagogical approach that integrates game principles and mechanics into structured learning environments to support engagement, motivation, and achievement (Pan et al., 2021; Sailer & Homner, 2020). Unlike gamification, which merely includes external rewards, GBL situates learning within the game itself, using play or design as a medium for knowledge construction and reflection.

The effectiveness of GBL can be theoretically explained by self-determination theory (Ryan & Deci, 2000) and flow theory (Csikszentmihalyi, 1990), as it enhances intrinsic motivation by satisfying learners' needs for autonomy, competence, and relatedness through meaningful goals, feedback, and interactive experiences. These mechanisms explain why GBL sustains learners' attention and supports deeper engagement in the learning process (Lüking et al., 2023).

Empirical research across computer science, engineering, music, language learning, and health-related disciplines (Alhammad & Moreno, 2018; Lee et al., 2023; van Gaalen et al., 2021; Van Roy & Zaman, 2018) demonstrates that well-designed GBL enhances motivation, enjoyment, and knowledge retention while fostering problem-solving and adaptive thinking.

Recent studies further suggest that involving students as game designers promotes deeper learning. Designing games requires analysing, organising, and creatively applying content, fostering imagination and theoretical understanding (Romero et al., 2019; Zaw Oo et al., 2024). In sustainability education, students who designed games reported enhanced creativity, skill development, and stronger content engagement (Mercer et al., 2017).

Despite these benefits, student-led game design remains underexplored in health education. This gap highlights opportunities to expand GBL into participatory, co-creative approaches that empower learners as knowledge constructors and communicators, providing a foundation for DBL in health promotion education.

Collaborative learning and creativity in higher education

Collaborative learning is an instructional approach where students work in small groups to solve problems, complete tasks, or create products and has been shown to enhance academic performance and teamwork-related competencies in higher education (Jaiswal et al., 2021). Rooted in sociocultural theory, collaborative learning conceptualises knowledge as socially constructed through interaction and mutual exchange, supporting cognitive and motivational development (Lenkauskaitė et al., 2020).

Collaborative creativity extends this foundation, focusing on co-constructing novel and valuable ideas through social interaction. Creativity, defined as the ability to produce original ideas and solve problems, thrives in open, reflective, and nonlinear environments (Glăveanu & Beghetto, 2021; Sternberg, 2009). As innovation and collaboration are core workplace competencies, fostering collaborative creativity through project-based learning is crucial for linking theory with practice (Baruah & Paulus, 2019; Thornhill-Miller et al., 2023).

Recent studies illustrate that collaborative creativity thrives in social, exploratory, and task-driven environments (Barrett et al., 2021). Compared with individual work, group collaboration generates more diverse ideas and encourages ideation, reflection, and shared knowledge transformation (Kim et al., 2019; Stolaki et al., 2023).

Game-based approaches further enhance collaborative creativity. While GBL casts students as players, involving them as game designers strengthens co-creativity and engagement (Romero et al., 2019). Designing games integrates knowledge application, idea expression, and problem-solving as core skills for collaboration and creativity (Mercer et al., 2017; Zaw Oo et al., 2024). Lee et al. (2023) found that students collaboratively designing games about learning disabilities demonstrated strong engagement, trust, and cohesion, underscoring the pedagogical value of co-design.

Method

Course design and pedagogical framework

This study was conducted in the undergraduate course *Health Promotion Theory and Practice* at a Taiwanese university. The course prepared public health students for theoretical assessment and applied programme planning. Instructional goals included understanding health promotion theories and acquiring competencies in needs assessment, intervention design, and programme evaluation.

The learning design integrated two pedagogical frameworks: the educational principles of health promotion (Naidoo & Wills, 2016) and Romero et al.'s (2019) iterative, needs-based game design model rooted in design thinking. The first framework guided the content structure, ensuring students progressed from understanding theories to applying them through needs assessment, intervention design, and evaluation. The second framework informed the design process, guiding students to iteratively analyse user needs, prototype, test, and refine their educational board games. Together, these frameworks aligned theoretical and procedural learning, linking health promotion competencies with creative, user-centred problem-solving relevant to real-world contexts.

Guided by these frameworks, the semester was structured into three phases (Figure 1). The first phase (Weeks 1–9) focused on theoretical foundations through lectures and discussions on health promotion principles, behaviour theories, and global strategies. The second (Weeks 10–11) developed planning skills through needs assessments and intervention design. The final phase (Weeks 12–18) centred on co-creative application, where students worked in groups of 6–7 to design educational board games for elementary students. Each group selected a specific health theme (e.g. balanced diet, infectious disease prevention, air pollution prevention), conducted audience analysis, defined learning objectives, and created

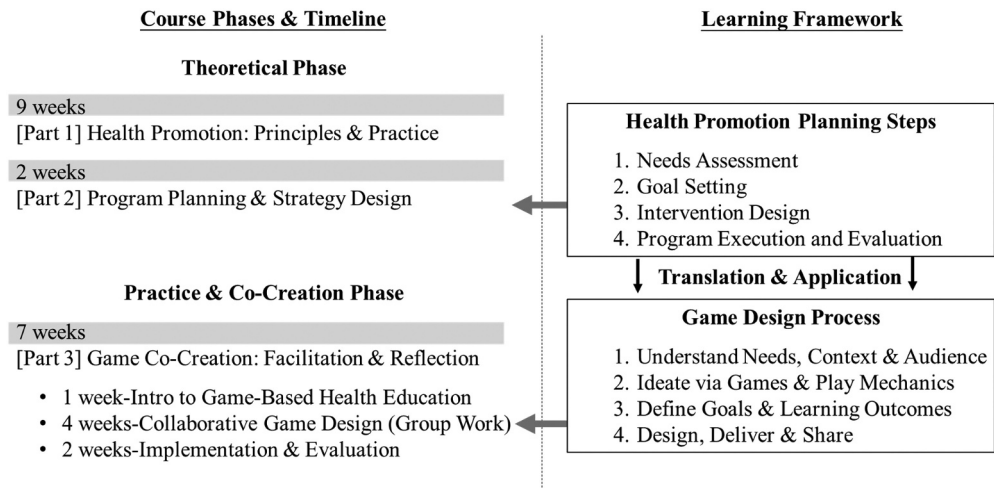


Figure 1. Health promotion course structure and GBL integration.

game mechanics and storylines to communicate health messages. Weekly workshops and consultations supported prototype development, peer playtesting, and iterative refinement, concluding with a public presentation that connected theory to community practice.

A guest expert in game-based education, along with a pre-developed learning handbook served as key instructional supports to enhance the co-creative learning process. The expert, a manager at a Taiwanese board game development company with expertise in educational game design and teacher training, provided practical insights into educational game mechanics and real-world implementation strategies, bridging theory with practice. Complementing this, the instructor prepared a learning handbook to scaffold student collaboration and design thinking throughout the project. Research indicates that inadequate teamwork can undermine the effectiveness of co-creative game-based projects, highlighting the need for explicit guidance in communication and coordination strategies (Lee et al., 2023). The handbook was organised into four main modules corresponding to the course phases: (1) teamwork and task allocation, (2) health promotion planning, (3) revision and preparation, and (4) implementation and reflection. Each module provided step-by-step guidance, templates, and examples to scaffold collaboration and game design for health education (Appendix A). In addition, task-specific checklists were provided to monitor progress and ensure alignment with the intended intervention logic. Together, these supports strengthened students' conceptual understanding, teamwork, and design coherence.

The course was grounded in a constructivist pedagogical philosophy, viewing learning as active, contextual, and socially mediated (Biggs & Tang, 2011; Vygotsky & Cole, 1978). By embedding co-creation and design thinking (Brown, 2009), students acted as both educators and designers, supporting deep learning, autonomy, and creativity within participatory, learner-centred health promotion education.

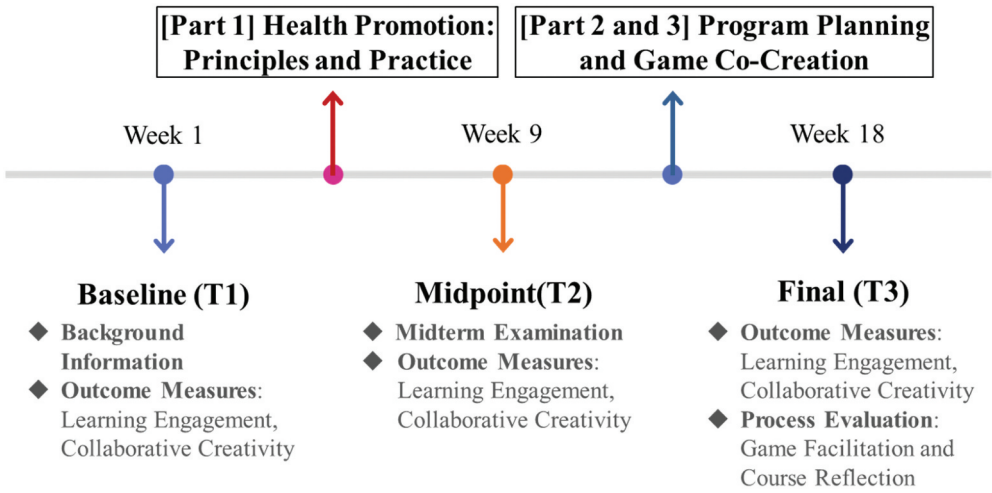


Figure 2. Timeline of instructional phases and data collection points.

Study design

Study design and context

This study employed a one-group repeated-measures design to evaluate the effects of the co-creative GBL intervention on student learning engagement and collaborative creativity. The intervention was implemented in an 18-week undergraduate course in health promotion, with data collected at three points – pre-test (T1), mid-test (T2), and post-test (T3). Between T1 and T2, students received conventional lectures; between T2 and T3, they participated in a six-week design-based intervention involving the collaborative creation of a health education board game. This structure enabled within-subject comparisons across instructional phases (Figure 2).

Participants

The study recruited 42 third-year public health students, plus three from related departments. All were enrolled in the same course and received identical instruction. Participation was voluntary and integrated into coursework. Survey data were de-identified after matching and had no impact on course grades. Ethical approval was granted by the Human Research Ethics Committee of National Cheng Kung University (Approval number: NCKU HREC-E-113-513-2; August 12, 2024). Informed consent was obtained from all participants. Students were informed that the study evaluated learning experiences before and after the course, while withholding the specific hypotheses to minimise expectancy bias.

Data collection

Quantitative and qualitative data were collected to assess two outcomes: learning engagement and collaborative creativity. A structured online questionnaire was administered at T1, T2, and T3 to capture students’ experiences before, during, and after the intervention, along with background variables for subgroup analyses. Survey instruments and reflection prompts are provided in Appendix B.

Background information

Students provided demographic and experiential information for T1, including gender, academic standing from the previous semester (top, middle, or bottom third of the class; students who were unsure or preferred not to answer could select 'Unknown/Unable to answer'), learning motivation towards the course (high, moderate, or low), experience with team-based projects in university (extensive, moderate, limited, or none), and perceived teamwork experience (mostly positive, neutral, or mostly negative). These items profiled students' background characteristics and prior collaborative experiences.

Learning engagement

Learning engagement was measured using a revised version of Generic Student Engagement Scale (GSES; Li et al., 2023). Eleven items from three subdomains were used: emotional engagement (4 items), peer and teacher interaction (4 items) and learning enjoyment (3 items). All the items were rated on a 5-point Likert scale and administered at T1, T2, and T3.

To ensure contextual alignment, item stems were adapted to match the learning phases. At T1 and T2, the students responded based on general experiences in undergraduate coursework (e.g. 'I am usually curious about the content of the courses I take'). At T3, the same items were reworded to reference the co-creative learning experience in the course (e.g. 'I felt curious about the content of this course'). This temporal adaptation enabled valid within-subject comparisons while maintaining conceptual continuity.

Collaborative creativity

Collaborative creativity was assessed using the Assessment Scale for Creative Collaboration (ASCC; Mavri et al., 2020), with 21 items across three subscales: Synergistic Social Collaboration (9 items), Distributed Creativity (7 items) and Time Regulation, and Achievement (5 items). The items were rated on a 5-point Likert scale and administered at T1, T2, and T3.

As with the engagement measure, item stems were adapted to the respective phase of the course. T1 and T2 items referred to group project experiences (e.g. 'In previous group work, most members of my team were interested in the task'), while T3 items targeted the current semester's collaborative project (e.g. 'In this semester's group work, most members of my team were interested in the task'). Instructions were tailored to distinguish general and course-specific responses. The three subscales were operationalised as follows:

- (1) Synergistic Social Collaboration evaluated affective and social cohesion, such as trust and reciprocity (e.g. 'There was a sense of belonging in my team').
- (2) Distributed Creativity assessed the extent to which creative contributions were shared and iteratively developed (e.g. 'Our team generated new and creative ideas for the educational game').
- (3) Time Regulation and Achievement measured time management and task completion (e.g. 'We managed time efficiently to meet deadlines').

Instrument validation

All the questionnaire items were reviewed by six experts with backgrounds in health promotion and teaching practice. A pre-test was conducted with 50 students from other cohorts to evaluate clarity and item performance. Content validity was assessed using a 4-point relevance scale and item-level content validity indices (I-CVI) ranged from 0.83 to 1.00, indicating strong expert agreement. Internal consistency reliability was calculated using Cronbach's alpha. The learning engagement scale demonstrated good reliability ($\alpha = 0.838$). The overall scale for collaborative creativity exhibited excellent reliability ($\alpha = 0.953$), with its three subscales ranging from $\alpha = 0.813$ to 0.942, supporting the internal consistency of all the constructs.

Additional measures

Students' midterm scores assessed their theoretical understanding, and end-of-semester reflections provided qualitative insights into engagement and perceived educational value. All data were processed confidentially. Student IDs were used only for matching surveys and scores, with de-identification handled by a research assistant uninvolved in instruction. After matching, all identifiers were removed, ensuring compliance with institutional ethics.

Data analysis

Quantitative data were analysed using SPSS 22.0 with $\alpha = 0.05$. Descriptive statistics summarised demographic characteristics and outcome variables. Repeated-measures ANOVA examined time effects (T1–T3) on engagement and creativity, and interactions between time and background variables (gender, academic standing, motivation, teamwork experience, midterm score). Post hoc tests were applied where interactions were significant. Effect sizes (partial η^2) followed Cohen's (2013) criteria: 0.01 small, 0.06 medium, 0.14 large.

Qualitative reflections were thematically analysed to complement quantitative results. A coding framework, derived from research questions and refined iteratively, guided theme identification of learning gains, team dynamics, and pedagogical value, providing contextual insights that triangulated the statistical findings.

Results

Participant characteristics

A total of 38 students completed all three waves of data collection (T1, T2, and T3), resulting in a 90.48% response rate. The majority were female (68.42%). Self-reported academic standing from the previous semester was evenly distributed: 28.95% identified as top-third performers, 31.58% as middle-third, and 23.68% as lower-third. Most students indicated moderate learning motivation towards the course (89.47%), while 78.95% reported having participated in more than five team-based learning activities. Furthermore, 68.42% described their perceived teamwork experience as mostly positive. These background characteristics suggest that the sample possessed a favourable disposition towards teamwork and course participation (Table 1).

Table 1. Participant demographic characteristics.

Variable	Category	Frequency	Percentage
Gender	Male	12	31.58
	Female	26	68.42
Self-rated academic standing (Previous semester)	Top third	11	28.95
	Middle third	12	31.58
	Bottom third	9	23.68
	Unknown/Unable to answer	6	15.79
Course learning motivation	High (Very interested, actively engaged)	2	5.26
	Moderate (Somewhat interested, open to learning)	34	89.47
Team project experience in university	Low (Not interested, learning reluctantly)	2	5.26
	Extensive (more than 5 times)	30	78.95
	Moderate (3–5 times)	8	21.05
Perceived teamwork experience	Mostly positive	26	68.42
	Neutral	10	26.32
	Mostly negative	2	5.26

Main effects of time on learning outcomes

Significant time effects were observed for learning engagement and collaborative creativity (Table 2). For learning engagement, mean scores increased steadily from T1 ($M = 3.53$, $SD = 0.39$) to T2 ($M = 3.58$, $SD = 0.45$), and to T3 ($M = 3.88$, $SD = 0.48$). The analysis revealed a significant main effect of time ($F[2, 74] = 20.580$, $p < 0.001$, partial $\eta^2 = 0.357$). Post hoc comparisons confirmed that engagement at T3 was significantly higher than at T1 ($p < 0.001$) and T2 ($p < 0.001$), suggesting that the co-creative learning experience led to substantial improvement in student engagement.

Collaborative creativity exhibited a significant increase over time. Mean scores rose from T1 ($M = 3.71$, $SD = 0.39$) to T2 ($M = 3.76$, $SD = 0.46$) to T3 ($M = 4.00$, $SD = 0.55$). The main effect of time was statistically significant ($F[2, 74] = 10.394$, $p < 0.001$, partial $\eta^2 = 0.219$). Post hoc comparisons indicated significant improvements between T1 and T3 ($p = 0.001$) and between T2 and T3 ($p = 0.003$), reflecting consistent growth throughout the intervention period.

Table 2. Time effects on learning outcomes (repeated measures ANOVA).

	Time point	MEAN $\pm$ SD	F	p	Partial η^2	Bonferroni pairwise comparison
Learning engagement	T1	3.53 $\pm$ 0.39	20.580	< 0.001	0.357	T3 > T1 ($p < 0.001$)
	T2	3.58 $\pm$ 0.45				T3 > T2 ($p < 0.001$)
	T3	3.88 $\pm$ 0.48				T1 vs T2: n.s.
Collaborative creativity	T1	3.71 $\pm$ 0.39	10.394	< 0.001	0.219	T3 > T1 ($p = 0.001$)
	T2	3.76 $\pm$ 0.46				T3 > T2 ($p = 0.003$)
	T3	4.00 $\pm$ 0.55				T1 vs T2: n.s.
Synergistic social collaboration	T1	3.68 $\pm$ 0.49	8.583	< 0.001	0.188	T3 > T1 ($p = 0.001$)
	T2	3.79 $\pm$ 0.53				T3 > T2 ($p = 0.020$)
	T3	4.04 $\pm$ 0.64				T1 vs T2: n.s.
Distributed creativity	T1	3.88 $\pm$ 0.40	2.379	0.100	0.060	–
	T2	3.79 $\pm$ 0.49				
	T3	3.94 $\pm$ 0.52				
Time regulation and achievement	T1	3.53 $\pm$ 0.49	15.634	< 0.001	0.297	T3 > T1 ($p < 0.001$)
	T2	3.66 $\pm$ 0.53				T3 > T2 ($p < 0.001$)
	T3	4.03 $\pm$ 0.60				T1 vs T2: n.s.

Mean $\pm$ SD refers to the arithmetic mean and standard deviation. Partial η^2 represents effect size, interpreted according to Cohen's (2013) guidelines: 0.01 = small, 0.06 = medium, 0.14 = large. n.s. = not significant. T1 = Pre, T2 = Mid, T3 = Post.

Subscale-level changes in collaborative creativity

Among the three subscales, Synergistic Social Collaboration demonstrated significant improvement, with scores increasing from T1 ($M = 3.68$, $SD = 0.49$) to T3 ($M = 4.04$, $SD = 0.64$). This represented a large effect size ($F[2, 74] = 8.583$, $p < 0.001$, partial $\eta^2 = 0.188$). Time Regulation and Achievement exhibited substantial gains, rising from T1 ($M = 3.53$, $SD = 0.49$) to T3 ($M = 4.03$, $SD = 0.60$), with a significant effect of time ($F[2, 74] = 15.634$, $p < 0.001$, partial $\eta^2 = 0.297$).

However, Distributed Creativity exhibited only a slight increase from T1 ($M = 3.88$, $SD = 0.40$) to T3 ($M = 3.94$, $SD = 0.52$), which was not statistically significant ($F[2, 74] = 2.379$, $p = 0.100$, partial $\eta^2 = 0.060$). This suggested that while improvements were observed in social collaboration and task management, the distribution of creative contributions among the team members remained inconsistent.

Interaction effects by background variables

To examine whether students' background characteristics influenced their intervention response, repeated-measures ANOVA with interaction terms was conducted for six variables: gender, academic standing, learning motivation, teamwork experience, perceived teamwork experience, and midterm performance. This identified subgroups that exhibited differential changes across time (Table 3).

For learning engagement, a significant main effect of time was found ($F[2, 74] = 5.531$, $p = 0.007$, partial $\eta^2 = 0.175$), indicating overall improvement after the intervention. However, none of the time X background variable interactions reached statistical significance, suggesting consistent gains across demographic and motivational profiles.

Conversely, significant interaction effects were observed for collaborative creativity. Time X academic standing interaction reached significance ($F[4, 70] = 3.365$, $p = 0.007$, partial $\eta^2 = 0.288$), with students in the lower academic tier exhibiting significant increases between T1 and T2 ($p = 0.045$) and between T1 and T3 ($p = 0.001$). Similarly, a significant time X learning motivation interaction was found ($F[4, 70] = 3.661$, $p = 0.011$, partial $\eta^2 = 0.227$), with low-motivation students reporting greater gains from T2 to T3 ($p = 0.002$) and T1 to T3 ($p = 0.005$).

These findings indicate that while learning engagement improved broadly across the subgroups, collaborative creativity gains varied depending on students' academic and motivational profiles.

Qualitative reflections on co-creative GBL

Open-ended reflections collected at the end of the course revealed three key themes. First, students reported that the co-creation task helped them transform abstract theories into applied practice. One student noted, 'Designing a game made me realise how difficult it is to simplify complex health messages'. Several students expressed that the task deepened their understanding as it required them to translate theoretical content into child-friendly formats. Second, the activity enhanced collaborative competencies. The iterative nature of the task encouraged peer interaction, idea sharing and joint decision-making. One participant commented, 'We learned to value everyone's strengths.

Table 3. Time X background variable interaction effects on learning outcomes (repeated measures ANOVA).

Outcome variable	Source	F	p	Partial η^2	Bonferroni pairwise comparison
Learning engagement	Time (main effect)	5.531	0.007	0.175	T3 > T1 ($p < 0.001$) T1 vs T2: n.s. T2 vs T3: n.s.
	Time X Gender	0.170	0.844	0.006	–
	Time X Academic standing	1.066	0.395	0.110	–
	Time X Learning motivation	0.279	0.890	0.021	–
	Time X Teamwork experience	2.386	0.102	0.084	–
	Time X Perceived teamwork experience	0.079	0.988	0.006	–
	Time X Midterm exam score	1.157	0.341	0.082	–
	Time (main effect)	2.749	0.083	0.180	–
	Time X Gender	0.624	0.544	0.048	–
	Time X Academic standing	3.365	0.007	0.288	Lower-ranking group T3 > T1($p = 0.001$) T2 > T1($p = 0.045$) T2 vs T3: n.s.
Collaborative creativity	Time X Learning motivation	3.661	0.011	0.227	Low-motivation group T3 > T1($p = 0.005$) T3 > T2($p = 0.002$) T1 vs T2: n.s.
	Time X Teamwork experience	2.292	0.122	0.155	–
	Time X Perceived teamwork experience	2.546	0.051	0.169	–
	Time X Midterm exam score	1.636	0.180	0.116	–

T1 = Pre, T2 = Mid, T3 = Post. n.s. = not significant. Partial η^2 represents effect size, interpreted according to Cohen's (2013) guidelines: 0.01 = small, 0.06 = medium, 0.14 = large.

The success of the game came from combining different ideas'. Third, students developed greater awareness of audience needs. Designing for elementary school students increased their sensitivity to learner perspectives and sense of professional identity. 'The moment our game clicked with the kids, I felt like a real health educator', a student reflected. These findings suggest that the co-GBL approach fostered cognitive engagement, teamwork, and real-world relevance in public health learning.

Discussion

This study examined the effects of a co-creative GBL intervention on two key outcomes – student engagement and collaborative creativity – while exploring potential variations across students with different academic standings and motivational levels. These aims align with the study's two research questions, both of which were supported by the findings. The results confirmed significant overall improvements in engagement and collaborative creativity, providing empirical support for the educational value of co-creative GBL in health promotion education. These results extend prior research on the motivational and cognitive benefits of game-based pedagogies (Zhang & Yu, 2022), and highlight the additional value of positioning students as game designers rather than solely as players (Romero et al., 2019; Zaw Oo et al., 2024). This participatory approach, as emphasised by Lee et al. (2023) and Romero et al. (2019), fosters deeper learning by integrating metacognitive reflection, content transformation, and creative problem-solving.

Specifically, subgroup analyses indicated that while engagement gains were consistent across all groups, improvements in collaborative creativity were more

pronounced among students with lower academic performance or motivation. This suggests that co-creative GBL may particularly empower less confident learners by providing shared responsibility and tangible creative outcomes that enhance agency and participation. From a sociocultural perspective (Vygotsky & Cole, 1978), co-creation can be understood as constituting authentic, interaction-rich environments for joint meaning-making, aligning with Glăveanu and Beghetto's (2021) view of creativity as cultural participation. Co-creative GBL's concrete outputs, shared responsibilities, and peer interdependence may particularly benefit learners with limited participation in traditional settings, enhancing their agency and participation within a community of practice (Wenger, 1998). Similar benefits for such learners have been reported in structured collaborative learning (Gillies, 2016), with more recent studies in design-based and co-creative learning contexts highlighting comparable patterns of learner engagement and agency (Lee et al., 2023; Pan et al., 2021). In this study, audience-specific deliverables, such as educational board games for elementary students, acted as 'tangible anchors' (Brown, 2009), enabling the translation of abstract health concepts into actionable educational tools that strengthened students' ability to communicate complex health messages effectively.

The absence of improvement in the Distributed Creativity dimension diverges from earlier findings reported by Mercer et al. (2017), who reported equitable creative participation in other co-design contexts. Cultural and structural factors, including implicit hierarchy effects (Kim et al., 2019), may have limited balanced idea-sharing, highlighting the need for scaffolding strategies such as role rotation and structured brainstorming (Romero et al., 2019).

Beyond student outcomes, the process had implications for educator development. Furthermore, facilitating co-creative GBL required instructors to shift from content delivery to design facilitation, involving group scaffolding, conflict mediation, and equitable participation support. Such shifts align with calls for educators to strengthen facilitation and adaptive pedagogy skills, building institutional capacity for innovative, collaborative learning.

The observed improvements align with the Public Health Specialists Act (Yeh, 2024) and international health promotion frameworks (World Health Organization, 2022), particularly in communication, teamwork, and community-oriented intervention design. This study's co-creative GBL model can be framed as a four-stage instructional process (knowledge construction, needs assessment and audience analysis, collaborative design and iteration, and authentic context validation) that is adaptable to other applied disciplines integrating competency-based experiential learning.

Qualitative reflections further illustrate how co-creative GBL supports multidimensional engagement (Bond et al., 2020) and identity-in-practice (Cruess et al., 2019). Students reported (1) the challenge of translating theory into age-appropriate messages, (2) navigating creative conflicts to strengthen cohesion, and (3) enhanced professional identity through authentic audience interaction. These findings underscore the integrative potential of participatory design to foster cognitive, social, and identity-related outcomes, which remain underexplored in current health education literature.

Overall, the study addressed its primary objective by demonstrating that co-creative GBL can effectively enhance both engagement and collaborative creativity among health promotion students, with especially strong benefits for underachieving or less motivated

learners. These findings support the integration of participatory, design-based pedagogies into competency-oriented health education, where creativity and collaboration are critical professional outcomes. By linking these results with broader theoretical and policy frameworks, co-creative GBL emerges as both a pedagogical innovation and a scalable strategy for future-oriented health promotion education.

Limitations and future directions

This study has several limitations. First, the sample was limited to one course in a single institution. Additionally, the relatively small number of students in some subgroups (e.g. by academic standing or motivation level) constrains the generalisability of the findings. Second, the study did not include a control group. Therefore, the observed improvements in engagement and collaborative creativity cannot be attributed solely to the co-creative GBL intervention; similar gains might have emerged through other concurrent learning activities or natural course progression. Third, because the surveys were administered by the course instructors, students may have provided socially desirable responses despite assurances of confidentiality. Fourth, the wording of the engagement and collaborative creativity items was adapted across the three measurement points to ensure contextual relevance to each learning phase. Although this maintained conceptual consistency, it may have introduced slight variations in item interpretation over time.

Future research should employ more rigorous experimental designs (Sailer et al., 2017; Zhang & Yu, 2022), include control groups, and examine varied institutional and cultural contexts to test the robustness of these findings. Additionally, targeted interventions that promote distributed creativity (Barrett et al., 2021; Stolaki et al., 2023) and minimise response bias may further enhance the validity and equity of collaborative learning outcomes.

Conclusion

This study demonstrated the pedagogical value of integrating co-creative GBL into health promotion education, exhibiting significant improvements in both student engagement and collaborative creativity. The greatest gains were observed among students with lower academic standing or motivation, suggesting that the participatory, design-based process provided an inclusive pathway for learning. By engaging these students as co-designers rather than passive participants, the course fostered a sense of ownership, confidence, and peer interdependence.

The co-creative structure functioned as a social learning mechanism, where collaboration, iterative feedback, and shared decision-making enabled all learners to contribute meaningfully regardless of prior ability. Through tangible creation and reciprocal support, students with lower confidence demonstrated both competence and sustained motivation, accounting for their comparatively greater improvement. Moreover, the process transformed traditional classroom dynamics, encouraging reflection, communication, and mutual respect.

At a broader level, these findings highlight how co-GBL bridges theory and practice in health promotion education. It situates learning within authentic, creative problem-solving contexts that mirror real-world collaboration among health professionals. By

aligning with national competency-based reforms and global creativity-driven education frameworks, co-creative GBL offers an evidence-based model for inclusive, engaging, and future-oriented higher education.

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Ethical statement

The study protocol was reviewed and approved by the Human Research Ethics Committee of National Cheng Kung University (approval number: NCKU HREC-E-113-513-2, dated 12 August 2024).

Informed consent

Informed consent was obtained from all the participants before data collection.

Author contribution

LL Liao: Conceptualisation, Methodology, Data curation, Funding acquisition, Writing – original draft, Writing – review & editing. IJ Lai: Investigation, Methodology, Supervision, Writing – review & editing. LC Chang: Methodology, Formal analysis, Writing – review & editing. CC Hsu: Methodology, Formal analysis, Writing – review & editing. YP Lo: Methodology, Formal analysis, Writing – review & editing.

AI statement

The authors confirm that no generative artificial intelligence (AI) tools or AI-assisted technologies were used in the conception, drafting, writing, editing, or revision of this manuscript. All content, analysis, and interpretations are entirely the authors' own.

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